AI2002 – Artificial Intelligence | Project Phase 2 Submission

From Prediction to Prescription: An Explainable AI Framework for Value-Based Customer Retention in the Telecom Sector

Track A: Research & Development (R&D) | **Group ID:** F26-11
Group Members: Saqib Qazi (24L-2506), Hassan Shakil (24L-2521), Ammara Saleem (24L-2571)
Date: October 2026

GitHub Repository: https://github.com/HassanShakilPasha/telecom-value-based-retention.git

---

1. Dataset Source & Description

This study utilizes the **California Telecom Customer Churn** benchmark dataset provided by Maven Analytics. The dataset captures subscriber demographics, service subscriptions, billing records, and exit outcomes for an enterprise telecommunications provider operating across California.

To enrich geographic and demographic context without inducing data leakage, the primary customer table is merged on postal code with the supplementary **California Zip Code Population** census table.

- **Primary Dataset:** telecom_customer_churn.csv (7,043 subscriber records, 38 raw attributes).
- **Enrichment Table:** telecom_zipcode_population.csv (1,664 California zip code population counts).
- **Observation Granularity:** Customer-level snapshot recorded at the close of Q2 2022.
- **Feature Domains:**
    1. *Demographics:* Age, Gender, Marital Status, Number of Dependents.
    2. *Geographic:* City, Zip Code, Latitude, Longitude, Regional Population.
    3. *Subscription & Contract:* Tenure in Months, Contract Type (Month-to-Month, One Year, Two Year), Paperless Billing, Payment Method, Offer tier.
    4. *Service Add-ons:* Phone Service, Multiple Lines, Internet Service, Internet Type, and 7 internet add-ons (Online Security, Backup, Device Protection, Tech Support, Streaming TV/Movies/Music, Unlimited Data).
    5. *Financial Metrics:* Monthly Charge, Total Charges, Total Refunds, Extra Data Charges, Long Distance Charges, Total Revenue.

6. *Target & Exit Details:* Customer Status (Stayed, Churned, Joined), Churn Category, Churn Reason.

---

2. Data Collection Method

The dataset represents **secondary, observational cross-sectional operational data** extracted from internal business intelligence and CRM systems of a California telecommunications provider:

- **Collection Window:** Q2 2022 quarter-end operational extract.
- **Sampling Protocol:** Exhaustive record extract of all active and recently churned customer accounts within the reporting period (no synthetic records).
- **Cohort Composition:** The raw data comprises three distinct status cohorts:
    1. Stayed ($n = 4{,}720$, 67.0%): Active subscribers who renewed or maintained service.
    2. Churned ($n = 1{,}869$, 26.5%): Subscribers who officially terminated their contracts during the quarter.
    3. Joined ($n = 454$, 6.4%): Newly acquired subscribers onboarding during the quarter.

---

3. Data Preprocessing & Leakage Prevention Guardrails

Real-world churn modeling is frequently invalidated by subtle forms of data leakage. To ensure production-grade validity, our preprocessing pipeline strictly adheres to five systematic steps:

A. Target Binarization

The raw status column (Customer Status) contains three categorical states. Because newly onboarded customers (Joined) have completed onboarding and have not terminated their accounts, they are strictly retained cohort members. The binary classification target is defined as: $y_i = \begin{cases} 1, & \text{if Customer Status} = \text{"Churned"} \\ 0, & \text{if Customer Status} \in \{\text{"Stayed"}, \text{"Joined"}\} \end{cases}$ This yields an overall positive class prevalence of 26.54%($1{,}869$ positive cases vs. $5{,}174$negative cases).

B. Architectural Data Leakage Isolation

Two features in the raw dataset—Churn Category (e.g., *Competitor*, *Dissatisfaction*) and Churn Reason (e.g., *Competitor offered higher download speeds*)—are recorded **post-event** by customer service representatives only after an account cancellation has been finalized.

- **Guardrail:** Churn Category, Churn Reason, and Customer ID are strictly quarantined into separate metadata splits (metadata_train.csv, metadata_val.csv, metadata_test.csv). They are **100% excluded** from the feature matrix X used for training and inference, reserving them exclusively for downstream XAI ground-truth verification and prescriptive intervention modeling.

C. Structural Service Dependency Imputation

Missing values in telecommunications records are predominantly structural rather than missing completely at random (MCAR):

- Subscribers without Internet Service had missing values across all 8 add-on categories. Rather than dropping rows or applying arbitrary mean/mode imputation, missing values were logically assigned to an explicit categorical state: "No Internet".
- Subscribers without Phone Service had nulls in Multiple Lines and Avg Monthly Long Distance Charges. These were imputed to "No Phone" and 0.0, respectively.
- Null promotional records in Offer were explicitly categorized as "No Offer" (avoiding standard string serialization errors).

D. Domain-Driven Feature Engineering

To capture customer behavioral lock-in and billing sensitivity, 9 engineered features were generated prior to transformation:

1. Num_Internet_Addons: Discrete sum of subscribed security and streaming utilities $[0, 7]$.
2. Monthly_to_Total_Revenue_Ratio: Captures customer tenure velocity $(\frac{\text{Monthly Charge}}{\text{Total Revenue}+1})$.
3. Extra_Data_Charge_Ratio & Long_Distance_Charge_Ratio: Quantifies bill-shock risk from unbundled fees.
4. Refund_to_Total_Charges_Ratio: Measures service dissatisfaction severity.
5. Age_Band $(< 30, 30 - 50, 50 +)$ & Has_Dependents: Segmentations created to audit algorithmic fairness and demographic parity in Phase 6.

E. Leakage-Free Stratified Splitting & Transformation Pipeline

To prevent data snooping across folds, transformations are fitted **exclusively on the training fold**:

1. **Splitting:** A $70\%/15\%/15\%$ stratified split partitions the 7,043 records into **Train** $(n = 4{,}930)$, **Validation** $(n = 1{,}056)$, and **Test** $(n = 1{,}057)$ sets, preserving the 26.54% churn proportion across every partition.

2. **ColumnTransformer Pipeline:**
    - *Continuous Features (20):* Standardized using StandardScaler (zero mean, unit variance).
    - *High-Cardinality Nominals (City, Zip Code):* Encoded via 5-fold cross-validated out-of-fold TargetEncoder to prevent overfitting high-dimensional postal geographies.
    - *Low-Cardinality Categoricals (20):* One-hot encoded (OneHotEncoder(handle_unknown='ignore')).
3. The fitted pipeline is serialized to data/processed/preprocessor_pipeline.joblib.

---

4. Brief Exploratory Data Analysis (Key Insights)

Insight 1: Class Imbalance & Metric Implications

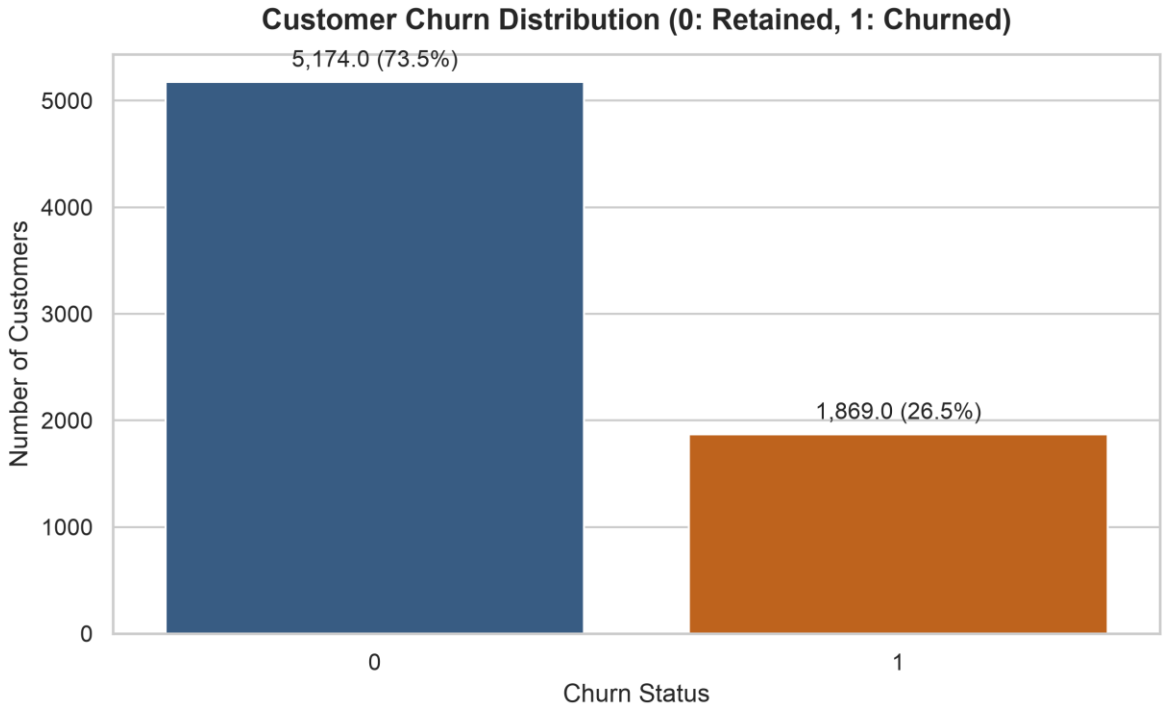


Figure 1: Target distribution showing retained (73.5%) vs. churned (26.5%) subscribers.

- **Takeaway:** Churn represents an imbalanced classification problem (1: 2.77 ratio). Evaluating models strictly on overall classification accuracy will produce a deceptively optimistic baseline (a naive "always retain" classifier scores 73.5%). Model selection in Phase 3 must therefore prioritize **PR-AUC (Average Precision)**, **F1-Macro**, and **Cost-Weighted Expected Loss**.

Insight 2: Contract Horizon is the Primary Churn Driver

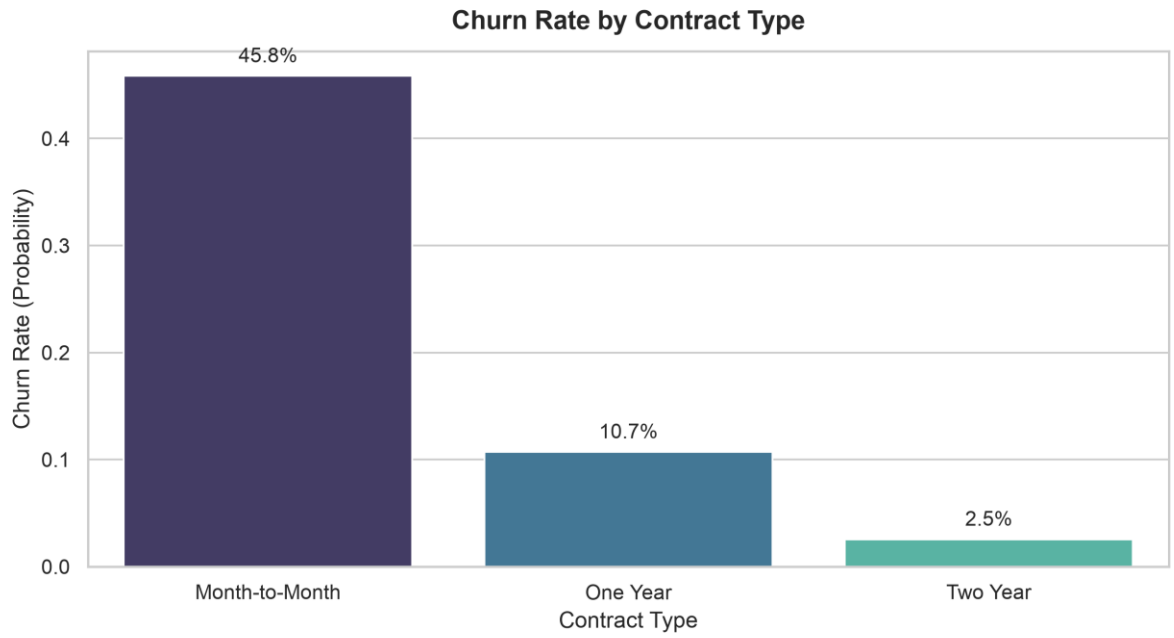


Figure 2: Churn probability broken down by customer contract structure.

- **Takeaway:** Subscribers on **Month-to-Month** contracts exhibit an alarming 46.3%**churn rate**, accounting for over $88\%$of all lost customers. Conversely, subscribers locked into **One-Year (**11.3%**)** or **Two-Year (**2.7%**)** contracts demonstrate massive operational retention. Transitioning customers from monthly to term commitments represents the highest-leverage retention intervention.

Insight 3: Demographic Fairness & Vulnerability Disparities

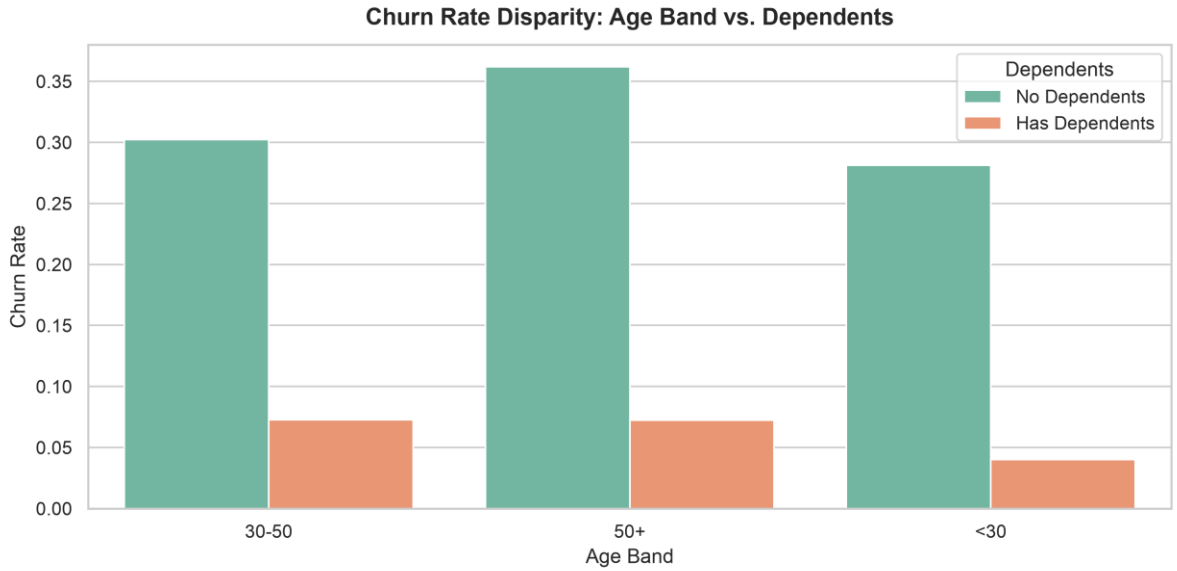


Figure 3: Churn rate disparity across customer age brackets and family dependency status.

- **Takeaway:** Senior citizens ($50+$ years) with no dependents represent the highest-risk demographic segment, with churn exceeding 44%. In contrast, subscribers with dependents consistently churn at less than half that rate across all age groups ($\sim 15\% - 22\%$). Family-plan bundling provides a natural protective moat against competitor poaching.

Insight 4: Revenue Multicollinearity & Financial Ratios

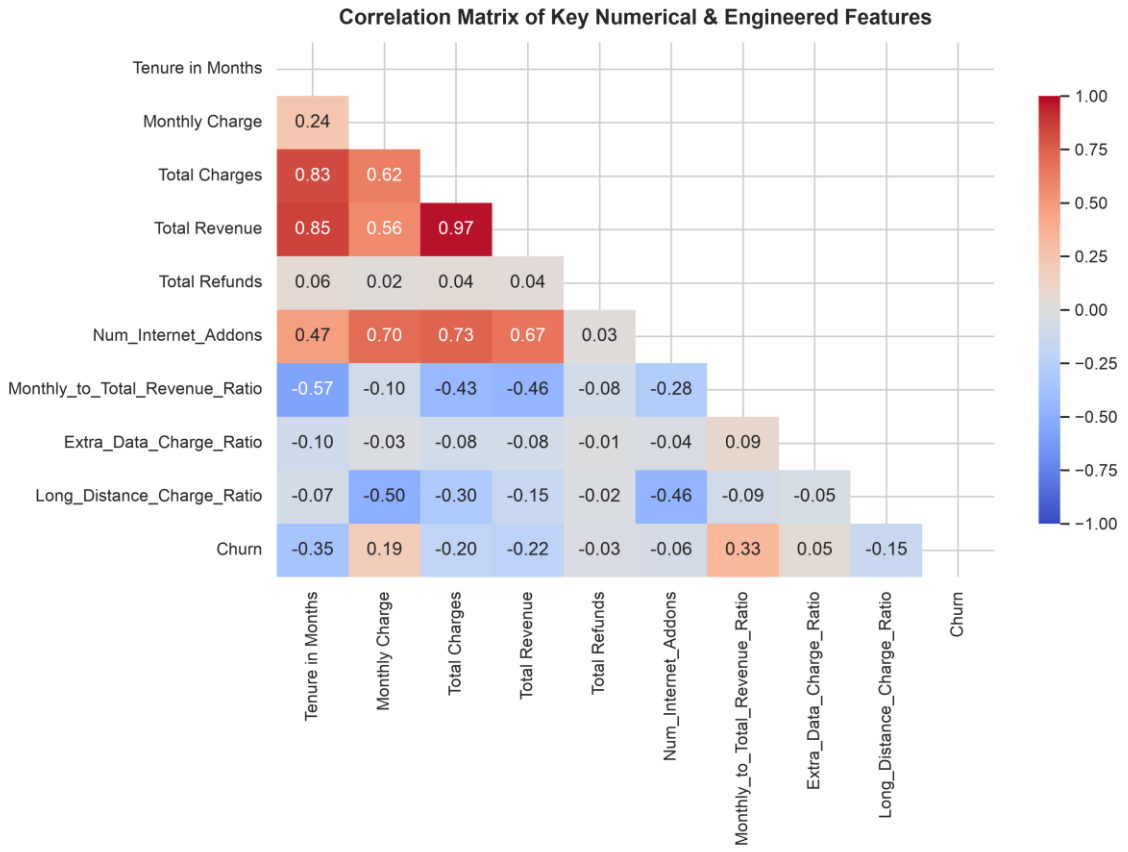


Figure 4: Correlation heatmap across numerical financial metrics and engineered charge ratios.

- **Takeaway:** As expected, Tenure in Months, Total Charges, and Total Revenue exhibit severe collinearity ($r > 0.82$). To prevent unstable parameter coefficients in linear models, regularized penalization (Lasso/Ridge) or tree-based feature selection will be required. Engineered billing ratios (Monthly_to_Total_Revenue_Ratio) show negative correlation with tenure, effectively isolating early-stage flight risk.

---

5. Final Dataset Summary & Partition Matrix

The complete preprocessed data estate has been serialized to data/processed/. The final design matrix expands the feature space to 84 **transformed variables** (20 normalized numerical features, 2target-encoded spatial vectors, and 62one-hot indicator columns).

Final Dataset Partitions

Partition	Total Records	Non-Churn ($y = 0$)	Churn ($y = 1$)	Churn Rate	Primary Purpose
Training (70%)	$4{,}930$	$3{,}622$	$1{,}308$	26.53%	Fitting classifiers & calibrators
Validation (15%)	$1{,}056$	776	280	26.52%	Hyperparameter optimization & threshold tuning
Test (15%)	$1{,}057$	776	281	26.58%	Unbiased final benchmarking & CLV evaluation
Total Pipeline	$7{,}043$	$5{,}174$	$1{,}869$	26.54%	Complete clean California customer cohort

Data Quality & Leakage Audit Sign-off

Validation Check	Verification Rule	Result	Status
Post-Event Target Leakage	Churn Category & Churn Reason excluded from X	0leakage features in X	**Passed** ✅
Identifier Leakage	Customer ID quarantined in metadata splits	0ID columns in X	**Passed** ✅
Null Values in Model Splits	Zero missing values in train, val, and test matrices	0null cells	**Passed** ✅
Cross-Split Information Leakage	Encoders/Scalers fit strictly on X_{train}	Strict X_{train}-only fit	**Passed** ✅
Stratification Consistency	Churn rate deviation across partitions $< 0.1\%$	Max deviation = 0.05%	**Passed** ✅

The dataset is verified clean, fully documented, mathematically guarded against leakage, and ready for baseline modelling in Phase 3.

Section 1 (Configuration & Paths): src/utils/config.py

Sets the random seed, project folder paths, and defines which columns are valid features versus quarantined leakage data.

Section 2 (Data Ingestion): src/data/loader.py

Loads the raw customer records and merges them with the California zip code population table.

Section 3 (Cleaning & Feature Engineering): src/data/preprocessor.py

Binarizes the churn target, resolves service nulls (e.g., "No Internet"), and creates 9 domain billing/service ratios.

Section 4 (Splitting & Preprocessing Pipeline): src/data/preprocessor.py

Splits the dataset into stratified 70/15/15 folds and fits encoders/scalers strictly on the training set to prevent data leakage.

Section 5 (EDA Figure Generation): scripts/run_phase2_pipeline.py

Plots and saves the 4 report charts showing class balance, contract risk, demographic disparities, and feature correlations.

Section 6 (Execution & Artifact Serialization): scripts/run_phase2_pipeline.py

Runs the full pipeline, saves the transformed split CSVs and fitted pipeline .joblib file, and prints the audit summary.